



SECP3213: BUSINESS INTELLIGENCE
2025/2026– SEMESTER 2

PROJECT GUIDELINE

1. Project Overview

This project requires students to design and develop a complete Business Intelligence (BI) solution by integrating multiple datasets, performing data preparation using Alteryx Designer, and generating insightful visualizations using Tableau or Power BI. Students must demonstrate data integration, transformation, analysis, and decision-support capability.

2. Project Requirements

- Use **minimum THREE (3) datasets** (must be relevant and meaningful).
- Datasets can be from:
 - Open data portals (e.g., government, Kaggle)
 - Industry-related datasets
 - Simulated datasets (if justified)
- Clearly define a business problem or decision-making scenario.

3. Project Phases & Submission

 Phase 1: Introduction (Proposal Stage)

Submission Type: Document (5–8 pages)

Content Required:

- 1. Project Title**
- 2. Problem Background**
 - Business context
 - Importance of the problem
 - Expected impact
- 3. Objectives** – state at least 3 objectives.
- 4. Datasets Description**
 - Source of each dataset
 - Type of data (structured/unstructured)
 - Key variables/features
- 5. Proposed BI Approach**
 - How Alteryx and visualization tools will be used.
 - Describe data integration, data cleaning and data transformation methods.
 - Describe the analysis that plan to be conducted and state at least 5 most appropriate visual forms to perform the analysis.

Deliverable:

- Report (PDF)

Deadline: **Friday - 15th May 2026; 5.00pm**

📌 Phase 2: Data Workflow (Alteryx Designer)

Submission Type: Workflow + Documentation

Requirements:

- Design BI architecture and build a complete workflow using Alteryx Designer
- Must include:
 - Data input from multiple datasets
 - Data cleaning (missing values, duplicates)
 - Data transformation (join, filter, aggregation)
 - Data integration (combine datasets)
 - Feature creation (if applicable)
- Apply analytical techniques (e.g., trend analysis, segmentation, basic predictive logic if relevant)

Deliverables:

- Alteryx workflow file (.yxmd)
- Brief documentation (3–5 pages in PDP):
 - Screenshots of workflow with explanation
 - Tools used
 - Challenges faced

Deadline: Friday – 5th June 2026; 5.00 pm

📌 Phase 3: Visualization & Dashboard

Submission Type: Data Modelling, Dashboard and Explanation

Requirements:

- Use either:
 - Tableau or Microsoft Power BI
- Create an interactive dashboard
- Include:
 - At least 3–5 meaningful visualizations
 - Filters/slicers for interactivity
 - Clear storytelling (insight-driven)
 - Minimum two dashboards and one storyboard.
- Focus on:
 - Trends
 - Comparisons
 - Key performance indicators (KPIs)
 - Business insights

Deliverables:

- Dashboard file (.twbx / .pbix)
- Explanation (2–3 pages in PDP):
 - Screenshots of dashboard
 - Key insights
 - Business recommendations

Deadline: Friday, 19th June 2026; 5.00 pm

Phase 4: Final Report Compilation

Students must compile **ALL phases into ONE complete report.**

Structure of Final Report:

1. Introduction (Phase 1)
2. Methodology & Workflow (Phase 2)
 - Explain the process to perform the data integration, data cleaning and data transformation.
 - Take a screenshot of the workflow and explain the tools selected and the tools configurations to design the workflow.
 - Present the results obtained from each of the process.
3. Visualization & Insights (Phase 3)
 - Explain the goals of each chart and justify on the charting choice, data visualization best practices and the interactive element applied.
 - Include the screenshots for the dashboards and storyboard.
 - Provide the insight or the analysis and suggestion for solution that can be made by the organisation to solve the problem from the analysis presented in the dashboards and storyboard.
4. Discussion
5. Conclusion
6. References
7. Appendix (workflow screenshots, additional charts)

Deliverable:

- Final Report: PDF
- Alteryx Workflow: .yxmd
- Visualization File: .twbx or .pbix
- Folder naming format: **BI_Project_GroupName**

Deadline: Wednesday, 24th June 2026; 5.00 pm

4. Assessment Criteria

Component	Marks
Problem Definition & Dataset Selection	20%
Alteryx Workflow Quality	30%
Visualization & Dashboard	25%
Insights & Business Value	15%
Report Quality & Presentation	10%

5. Important Guidelines

- Ensure data consistency and integrity
- Avoid using overly simple datasets
- Demonstrate critical thinking and problem-solving
- All work must be **original**.
- Properly cite all data sources.